

COL215 – Digital Logic and System Design
Department of Computer Science & Engineering,

IIT Delhi Semester I, 2025-26

Lab Assignment 8 part-3

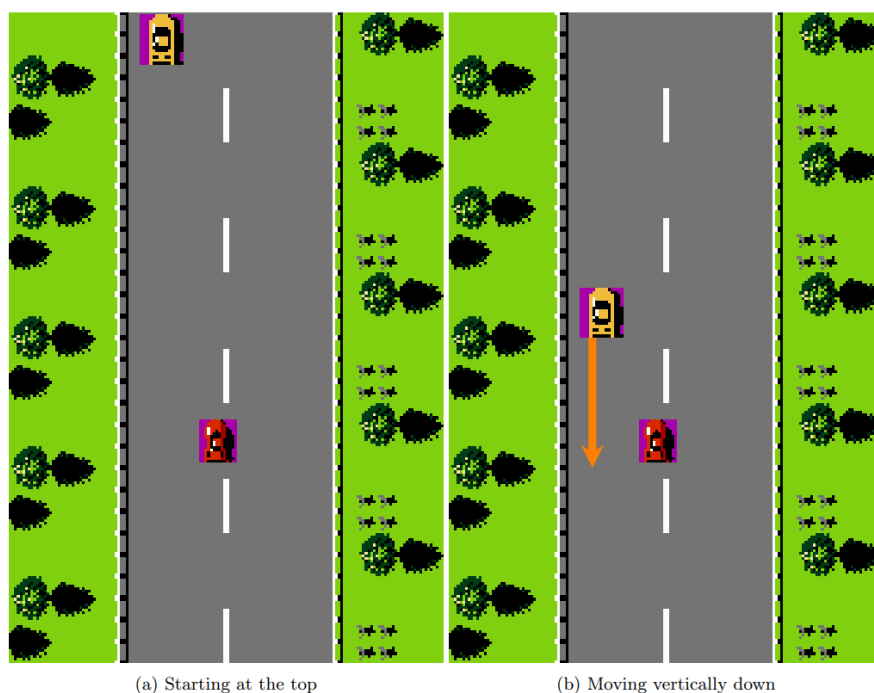
Lakshit Solanki Aakarsh Antriksh

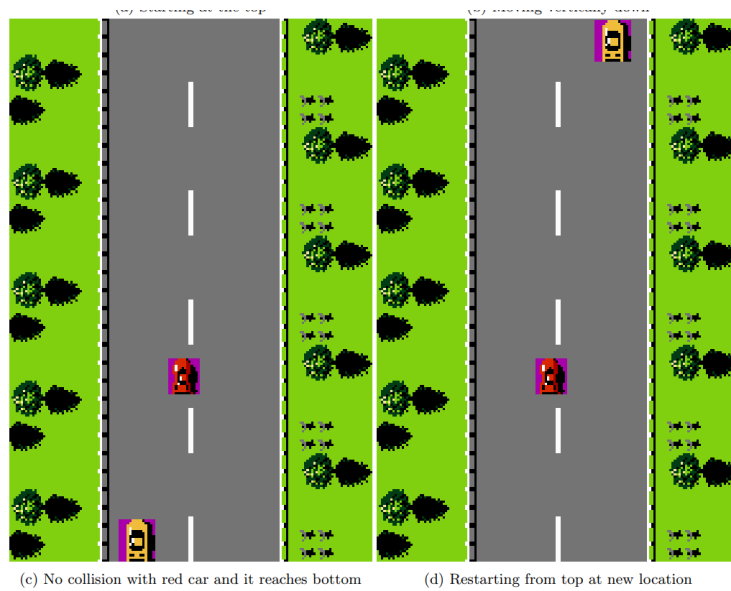
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Part III: Displaying a Rival Car

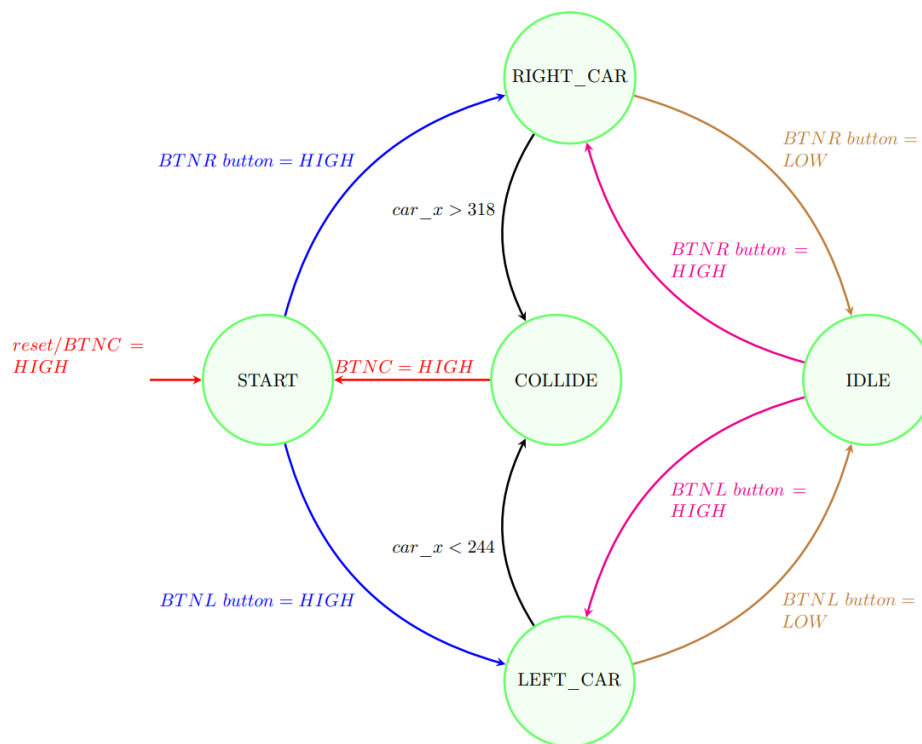
Tasks:

1. Implement a 8-bit pseudo random generator. Details are described in Section 2.3.1.
2. Create a single port ROM with the name `rival_car_rom`, with 12 bits width and 224 as depth of port A. Initialize it with `Rival_car_testing.coe` COE file.
3. Implement logic to randomly get horizontal pixel between 44 and 104 pixel using pseudo random generator and vertical pixel value fixed at (`OFFSET_BG_Y`). Refer to Figure 9 indicating the rival car starting position at the top.
4. Implement logic to move the rival car vertically down until it reaches the bottom boundary of the road.
5. Augment the FSM implemented to detect collision between main and rival car. Collision occurs when the hit boxes of both cars overlap.





The FSM:



The Collision state can also be reached by collision with the rival car.

Linear Feedback Shift Register (LFSR):

The **LFSR** generates an 8-bit pseudo-random sequence that repeats only after all possible non-zero states (255 cycles).

Each new random number is used to determine the **x-position** of the rival car when it spawns.

The seed is based on our entry numbers.

- The feedback polynomial depends on 3rd, 4th, 5th, and 7th bit of the dout register and works as the “shift” that changes the random sequence.
- The “en” enable sequence controls when random numbers are generated.
- Reset clears the dout back to SEED when the game restarts to keep the pattern deterministic.

Design Decisions-

Rival Car Spawning and Movement:

- **Rival Speed:** decides the number of cycles before the rival car is moved.
- **Rival_x_range:** Range of horizontal randomization.
- **Frame_end_tick:** Marks the start of each frame.

At each frame, if the rival car isn't on screen, it is spawned at the top with a randomised x-coordinate using the LFSR. Otherwise, the rival car is moved one pixel down. If the car reaches the bottom, it is set to inactive and so it disappears and is spawned in the next frame.

Collision Detection:

The system uses AABB(Axis Aligned Bounding Box) logic to detect overlapping regions. At every frame, the system checks for rival car collision by checking if there is an overlap in the area covered by the main car and the rival car. If there is an overlap, a collision is detected with a delay of one cycle.

```

wire rival_collision_detected = col_clk_q2;
always @(posedge pixel_clock) begin col_px_q <= collision_event_px; end
always @(posedge clk) begin {col_clk_q1, col_clk_q2} <= {col_px_q, col_clk_q1}; end

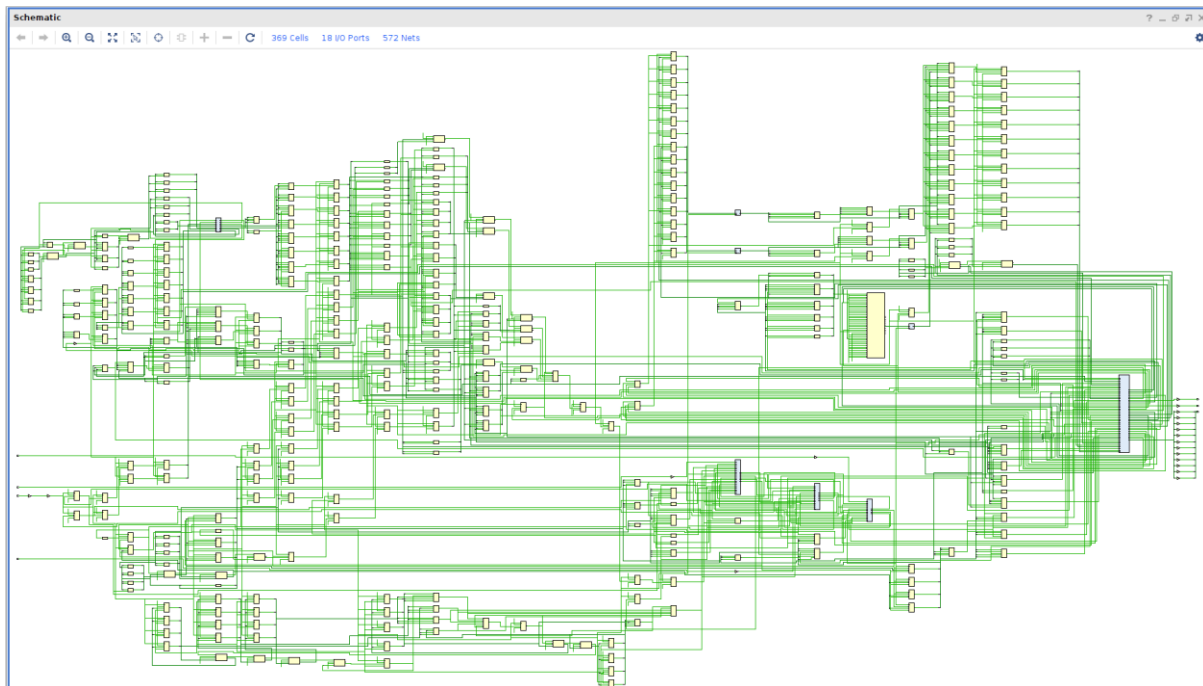
```

```

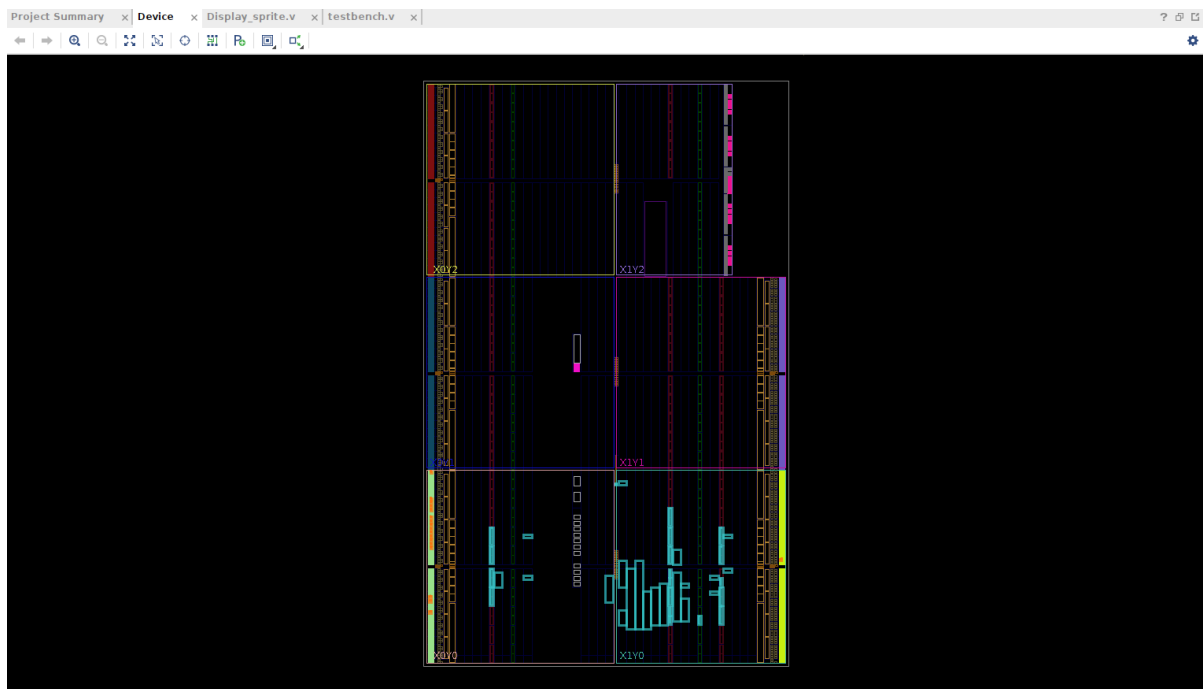
if (rival_active && state_px_q2 != START && state_px_q2 != COLLIDE) begin
    if ( (car_x_px_q2 < (rival_x + main_car_width)) &&
        ((car_x_px_q2 + main_car_width) > rival_x) &&
        (car_y_px_q2 < (rival_y + main_car_height)) &&
        ((car_y_px_q2 + main_car_height) > rival_y) )
    begin
        collision_event_px <= 1;
    end
end

```

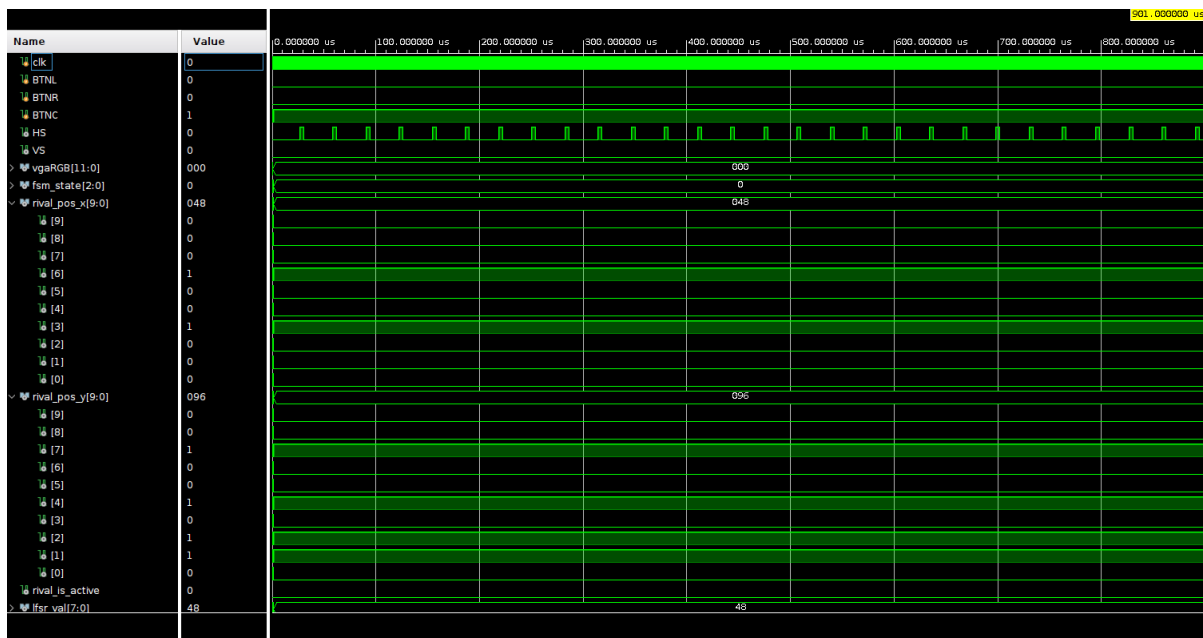
Schematics-



Synthesis Design-



Simulation



Resource Utilization-

1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs	382	0	0	20800	1.84
LUT as Logic	382	0	0	20800	1.84
LUT as Memory	0	0	0	9600	0.00
Slice Registers	255	0	0	41600	0.61
Register as Flip Flop	255	0	0	41600	0.61
Register as Latch	0	0	0	41600	0.00
F7 Muxes	0	0	0	16300	0.00
F8 Muxes	0	0	0	8150	0.00

3. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	14.5	0	0	50	29.00
RAMB36/FIFO*	12	0	0	50	24.00
RAMB36E1 only	12				
RAMB18	5	0	0	100	5.00
RAMB18E1 only	5				

* Note: Each Block RAM Tile only has one FIFO logic available and there

4. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	1	0	0	90	1.11
DSP48E1 only	1				

2. Slice Logic Distribution

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice	178	0	0	8150	2.18
SLICEL	130	0			
SLICEM	48	0			
LUT as Logic	382	0	0	20800	1.84
using O5 output only	2				
using O6 output only	264				
using O5 and O6	116				
LUT as Memory	0	0	0	9600	0.00
LUT as Distributed RAM	0	0			
LUT as Shift Register	0	0			
Slice Registers	255	0	0	41600	0.61
Register driven from within the Slice	186				
Register driven from outside the Slice	69				
LUT in front of the register is unused	33				
LUT in front of the register is used	36				
Unique Control Sets	18		0	8150	0.22

* * Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more in